CEMI Institution Classification Codebook

Version 1.0

A fully cited codebook for the 4-sector classification of Soviet R&D institutions in the CEMI Career Database

Companion to:

- *institution glossary (Nolting 4-sector classified, all sheets, classified).xlsx*
- *institution glossary (Nolting 4-sector classified, all sheets, unified).xlsx*

Built on the Nolting & Feshbach (1980, 1981) codification of the Soviet TsSU sectoral reporting framework, with the Graham (1975) three-pyramid model as historical antecedent. Variable template follows the structured 5-field block convention established in the historical-database codebook literature (Harvey & Press 1996, Ch. 5; Gil 2021, Ch. 3).

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Prepared for submission to Scientific Data

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1. Purpose and Scope

This codebook documents the classification of every R&D-performing organization listed in the CEMI Career Database glossary into one of five canonical codes: Academy, Branch, VUZ, Enterprise, and Other. The classification follows the four-sector decomposition codified by Nolting & Feshbach (1980; 1981) on the basis of the Soviet Central Statistical Administration (TsSU) sectoral reporting framework. Every coding decision is anchored in a verbatim, page-cited passage from the canonical English-language literature on Soviet science policy; every operational rule is auditable; and the variable template follows the structured 5-field block convention established in the historical-database codebook literature (Harvey & Press 1996, Ch. 5; Gil 2021, Ch. 3).

Scope. The codebook applies to all three sheets of institution glossary (Nolting 4-sector classified, all sheets, classified) – Sheet1 (Previous Institution, n = 445), Sheet2 (Part-Time Institution, n = 68), and Sheet3 (Transferred Institution, n = 121) – for a total of 634 institution records. The same coding logic is extended to data.xlsx (CEMI Personnel Data, 2,548 row-cell assignments).

2. Conceptual Foundations

2.1 Why the Nolting 4-sector framework

Three considerations make Nolting & Feshbach's 4-sector framework the most defensible classification spine for CEMI's institutional data:

(i) Direct reproduction of TsSU sectoral reporting. The four sectors are not Western analytic categories — they are the categories the Soviet statistical authority itself used. Nolting & Feshbach state this verbatim:

> [Q-1] *"It provides data on personnel engaged in conducting and supporting research and development (R & D) and on the distribution of R & D personnel by function, place of work, and branch of science. It also adjusts the Soviet data on R & D employment to make them comparable with U.S. data and compares the adjusted Soviet data and the U.S. figures."*
>
> — Nolting & Feshbach (1981), p. iii.
>
> [Q-2] *"The published distribution of scientific workers by major place of work covers the following broad categories: scientific institutions of branch ministries and other state agencies in charge of sectors of the economy (branch scientific institutions); scientific institutions under the academy system; VUZy; and a residual category consisting of industrial enterprises, nonresearch project and design organizations, state administrative staffs, and other organizations."*
>
> — Nolting & Feshbach (1981), p. 22.

(ii) Most extensive Western codification. The Nolting & Feshbach(1981) report runs to ~80 pages of definitions, tables, and an Appendix B that reproduces TsSU's own taxonomy in full. No competing English-language framework matches this coverage.

(iii) Adoption by downstream Soviet/Western scholarship. Cocks (1980) NSF report uses the same sectoral language; Kassel & Campbell (1980) RAND R-2533 structures its analysis around the same three-pyramid plus enterprise layer.

2.2 Historical genealogy

The 4-sector framework has deep antecedents. Imperial Russian science was already organized in three institutional frameworks; the Soviet system codified this as three pyramids by the 1930s; Western intelligence and academic literature converged on the same tripartite by 1959; and the fourth (enterprise) sector was added as Soviet statistics began separately reporting заводская наука in the 1970s.

Imperial Russian origin (Graham 1975):

[Q-2] *"In the first years of this century, scientific research in Russia was carried on within three different institutional frameworks: (a) the higher educational institutions, including the universities (ten in 1914) and the specialized higher schools in technological fields; (b) the Imperial Academy of Sciences, founded in 1725; (c) ministerial institutions promoting research in a few areas of national interest, such as geology, hydrography and agriculture."*

— Graham (1975), p. 306.

Soviet codification of three pyramids by the 1930s:

[Q-3] *"The system of scientific research that emerged in the Soviet Union by the 1930s was based on three pyramids: the Academy of Sciences of the USSR and the republican academies, the educational institutions, and the governmental ministries. In terms of intellectual eminence, there was no question which of the three pyramids enjoyed the commanding position."*

— Graham (1975), p. 327.

Centrality of the научно-исследовательский институт (NII) form:

[Q-4] *"In the Soviet Union today the term 'scientific research institute' (nauchno-issledovatel'skii institut) has a significance and a currency unequalled in any country in Western Europe or the Americas. The prototypical research institute in the Soviet Union is located in the Academy of Sciences of the USSR."*

— Graham (1975), p. 303.

Why Enterprise science was historically weak (Graham's explanation):

[Q-5] *"The Soviet planners originally paid very little attention to the development of on-site industrial laboratories, favouring instead the complex institutes under the administration of the industrial ministries."*

— Graham (1975), pp. 322–323.

Western tripartite consensus in 1959 (CIA):

[Q-6] *"The heart of scientific endeavor in the USSR is the Academy of Sciences, and its 150 Academicians stand in the front rank of Soviet science. The Academy, which is directly responsible to the Council of Ministers, has affiliates and associated academies in 13 of the 15 Union Republics… only about one-tenth of the scientific research done in the USSR is carried out within the Academy."*

— CIA (1959), § 4.

[Q-7] *"Higher Educational Institutions. Nearly half of all Soviet scientists are now employed in these institutions, which include some 39 universities and about 375 technical institutes."*

— CIA (1959), § 5.

[Q-8] *“Ministerial Institutes. Until the economic reorganization of 1957, research institutes under individual ministries employed about two-fifths of all Soviet scientists. These facilities, which ranged from plant laboratories to large research institutions, emphasized applied research in support of the industrial, military or other functions of each ministry.”*

— CIA (1959), § 6.

2.3 Anchoring in TsSU primary statistics

Nolting & Feshbach (1981) Appendix B reproduces the TsSU taxonomy of what is included in versus excluded from the 'science and science services' branch (наука и научное обслуживание). This is the authoritative reference for the boundary between Branch / VUZ / Academy / Enterprise and the Other residual.

[Q-9] *“I. Organizations under 'science and science services' branch. A. Scientific institutions (nauchnyye uchrezhdeniya): 1. NII's and their affiliates and divisions… 2. Independent scientific research laboratories. 3. Independent design organizations (KB's) conducting scientific research as well as design work. 4. Scientific and experimental stations… 5. Testing fields, supporting stations… 6. The U.S.S.R. Academy of Sciences, the republic academies of sciences, the specialized branch academies… 7. Organizations for preservation and study of fauna and flora… 8. Observatories. 9. Computer centers… 10. Museums… 11. State archives… 12. Libraries, book centers, and institutes of scientific and technical information.”*

— Nolting & Feshbach (1981), pp. 53–54.

3. The institution_type Variable (Structured 5-field template)

Following the structured-variable convention adopted in the historical-database codebook literature (Harvey & Press 1996, Ch. 5 on variable-block design; Gil 2021, Ch. 3 on per-variable documentation), every categorical variable is documented through five uniform fields: Type, Value Range, Level, Description, and Sources. The same 5-field block is used in the main CEMI Career Database codebook for variable documentation.

institution_type

Type	Character
Value Range	["Academy", "Branch", "VUZ", "Enterprise", "Other"]; optional sub-codes: "Mixed" (spans ≥ 2 sectors with documented evidence), "Don't know" (NA where sector cannot be determined from the available name).
Level	Institution (one row per Previous / Part-time / Transferred record in the CEMI glossary; one cell per personnel row in data.xlsx).
Description	Assigns each R&D-performing organization to one of the four canonical Soviet S&T sectors codified by Nolting & Feshbach (1979–1981) on the basis of TsSU reporting and Graham's (1975) three-pyramid typology; or to the Other residual for non-R&D bodies. Coded by descending-priority cascade R0–R7 documented in § 6 below.
Sources	Nolting & Feshbach 1980 (Science 207); 1981 (Foreign Economic Report Series P-95, No. 76); Nolting 1985 (Financing of R&D and Innovation in the U.S.S.R.); Cocks 1980 (NSF Science Policy in the USSR); Graham 1975; Kassel & Campbell 1980 (RAND R-2533); CIA 1959; USSR Council of Ministers Decree No. 760 of 1968-09-24; Decree of 1961-06-13 on scientific-pedagogical cadres

4. Sector Definitions, Evidence, and Coding Criteria

4.1 Academy (academic science / академический сектор)

Definition. The Academy sector comprises (i) the USSR Academy of Sciences (АН СССР) and its regional divisions; (ii) the 14 republic academies of sciences (RSFSR excluded — covered by USSR AS regional Divisions and Filials); (iii) the specialized branch academies — VASKhNIL (agriculture, under USSR Ministry of Agriculture), AMN (medicine, under USSR Ministry of Health), APN (pedagogy, under USSR Minvuz), USSR Academy of the Arts (Ministry of Culture).

Verbatim definitional evidence:

> **[Q-A1]** *"The scientific institutions under the academy system are engaged primarily in fundamental research; approximately 65 to 80 percent of the Soviet Union's fundamental research during the 1970's has been conducted by these institutions… The academy system is composed of three general sectors, each with its own research institutions: 1. The U.S.S.R. Academy of Sciences… 2. The academies of sciences of the separate republics… and 3. The specialized branch academies, subordinated to separate ministries, but supervised by the U.S.S.R. Academy."*
>
> — Nolting & Feshbach (1981), p. 27.
>
> **[Q-A2]** *"There are three types of academies — the all-union academy, republic academies, and branch academies. Subordinate to each academy are a number of institutes, laboratories, observatories, experimental stations, libraries, museums, and research ships. Most are organized around scientific or technical disciplines. As a rule, academy institutes tend to concentrate on fundamental research and generally constitute the leading Soviet facility in the particular scientific field."*
>
> — Cocks (1980), p. 47.
>
> **[Q-A3]** *"Reporting directly to the USSR Council of Ministers, the Academy has overall responsibility for development of R&D in natural and social sciences. In these areas, the Academy functions as a statewide center for the development of science policy."*
>
> — Cocks (1980), pp. 48–49.
>
> **[Q-A4]** *"The republic academies are dually subordinate to the USSR Academy and to their respective republic councils of ministers. Funding and administrative supervision are the responsibility of the councils of ministers; technical and functional supervision is provided by the USSR Academy."*
>
> — Cocks (1980), p. 50.
>
> **[Q-A5]** *"The Siberian Division of the Soviet Academy is unique in the Academy system… It is administratively subordinate to both the USSR Academy and the Council of Ministers of the Russian Republic (RSFSR). Funding is provided by the RSFSR, which has no republic academy of its own."*

— Cocks (1980), pp. 49–50.

[Q-A6] *"The Academy is unique in combining (a) national leadership in planning and coordination of S&T research with (b) status as a leading performer."*

— Kassel & Campbell (1980), p. 20.

Statistical magnitude:

Indicator	Value	Source
Share of scientific workers (1973)	9.0%	NF81 p. 27
Share of degree-holders (1973)	14.7%	NF81 p. 27
Share of official science expenditures	≈ 8%	Cocks 1980 p. 46
Share of fundamental research	65–80%	NF81 Ch. 3 p. 27; Cocks p. 46
Share of applied R&D	≈ 5%	Cocks 1980 p. 46
Internal split 1978 (USSR AS / Rep. / Branch AS)	40% / 41% / 19%	NF81 p. 27
Total institutions 1975 (USSR + Rep. + Branch AS)	246 + 393 + 228 = 867	NF81 Table 1, p. 4
Kassel & Campbell Academy R&D institute inventory	516 total (331 technology-relevant)	RAND R-2533, Table 3, p. 44

Sub-types: USSR Academy of Sciences (АН СССР) + Siberian / Far East / Urals Divisions + regional Filials (Bashkir, Dagestan, Karelian, Kazan, Kola, Komi); 14 republic academies; specialized branch academies (VASKhNIL, AMN, APN, USSR Academy of the Arts); within each: institutes (НИИ), laboratories, observatories, experimental stations, libraries, museums, research ships.

Coding criteria (Rule R2):

Code Academy if the institution name (Russian or English) contains any of: «АН», «Академия наук», «АМН», «ВАСХНИЛ», «АПН», «АН СССР», 'USSR Academy of Sciences', 'Academy of Sciences of [republic]', 'Academy of Medical Sciences', 'VASKhNIL', 'Academy of Pedagogical Sciences'. Override-to-VUZ exceptions: Academy of Foreign Trade, Academy of National Economy, Academy of MVD, Diplomatic Academy, Academy of Social Sciences (АОН при ЦК КПСС), Timiryazev Agricultural Academy, military VUZ-equivalent academies (Zhukovsky, etc.) — these are teaching/training academies, not the Academy of Sciences sector. Override-to-Branch exception: Academy of Municipal Economy named after Pamfilov (USSR Ministry of Housing and Communal Economy R&D body).

Worked examples from the classified glossary:

Sheet	Abbreviation	English name	Evidence
Previous Institution	АН Азербайджан. ССР	Academy of Sciences of Azerbaijan SSR	RU: 'АН '
Previous Institution	АН Казахской ССР	Academy of Sciences of Kazakh SSR	RU: 'АН '
Previous Institution	АН Литовской ССР	Academy of Sciences of Lithuanian SSR	RU: 'АН '
Previous Institution	АН Молдавской ССР	Academy of Sciences of Moldavian SSR	RU: 'АН '
Previous Institution	АН Таджиковой ССР	Academy of Sciences of Tajik SSR	RU: 'АН '

4.2 Branch (отраслевая наука / NO's)

Definition. The Branch sector comprises operationally independent R&D institutions — NII, TsNII, VNII, OKB, TsKB, SKB, KB, GSPI, project institutes, NPO — subordinated to industrial ministries, state committees (Gosplan, GKNT, Goskomstat, Gosbank, Gosstroy), main administrations (главки), and trusts. It is the modal sector for applied R&D and the largest employer of scientific workers in Soviet S&T.

Verbatim definitional evidence:

> **[Q-B1]** *"Branch scientific institutions are operationally independent institutions specializing in R&D pertinent to their own branch of economic or social activity. Most of them are subordinate to ministries in charge of industry, but some fall under ministries for other branches of the national economy, such as construction, transportation, health, and culture. Nearly half of all scientific workers (46.6 percent in 1973) are employed in branch scientific institutions… about 85 percent of official science expenditures are made by this sector."*
>
> — Nolting & Feshbach (1981), pp. 25–26.
>
> **[Q-B2]** *"Currently there are about 30 union-level industrial ministries in the USSR Council of Ministers. The research institutes and design organizations subordinate to the industrial ministries at the union and republic levels constitute what probably are the most important Soviet resources for applied R&D… the ministerial branch system includes half of all scientists and engineers in the Soviet Union and garners more than three-fourths of all the expenditures for R&D."*
>
> — Cocks (1980), p. 57.
>
> **[Q-B3]** *"To be sure, the branch sector of science is not just 'industrial.' There are other branches, such as construction, trade, and justice, though the bulk of this sector consists of industrial scientific organizations."*
>
> — Cocks (1980), p. 57.
>
> **[Q-B4]** *"The basic units of Soviet industry traditionally have been and, to a large extent, still are research institutes, design bureaus, and production enterprises. These were grouped by product line and subordinated to units of their respective ministries known as chief directorates or main administrations (Glavki)."*
>
> — Cocks (1980), p. 58.
>
> **[Q-B5]** *"Branch scientific institutions, that is, institutions subordinate to the ministries and other agencies in charge of branches of the national economy, increased by 78.6 percent from 1956 to 1975."*
>
> — Nolting & Feshbach (1981), pp. 3–4.

Statistical magnitude:

Indicator	Value	Source

Share of scientific workers (1973)	46.6%	NF81 pp. 25–26
Cocks alternative figure	≈ 58%	Cocks 1980 p. 46
Share of official science expenditures	80–85%	NF81 p. 26; Cocks p. 46
Share of applied R&D	90–95%	Cocks 1980 p. 46
Total branch scientific institutions 1956 → 1975	2,018 → 3,604 (+78.6%)	NF81 Table 1, p. 4
Kassel & Campbell ministry-system inventory	≈ 1,300 institutions	RAND R-2533, p. 8
Production associations (PO) by fall 1976	> 3,000	Cocks 1980 p. 71

Sub-types: NII (научно-исследовательский институт), TsNII (центральный НИИ — branch lead institute), VNII (всесоюзный НИИ — all-union), KB (конструкторское бюро — design bureau), OKB (опытно-конструкторское бюро — test/experimental design bureau), TsKB (центральное КБ), SKB (специальное КБ), PKB (проектно-конструкторское бюро), GSPI (государственный союзный проектный институт), NPO (научно-производственное объединение — combining an NII with design subdivisions and serial-production plant); project institutes; independent experimental/pilot plants (опытные заводы) not producing for sale.

The NPO/PO distinction (Decree No. 760/1968):

> **[Q-B6]** *"scientific–production associations, comprising research institutes with design–engineering and technological divisions and industrial enterprises, capable of creating in short periods the newest specimens of machines, equipment, instruments and other industrial products, of subjecting them to experimental verification at the level of the latest achievements of science and technology, of developing the technological tooling for their manufacture, and, upon completion of the development of designs and technology, of transferring the products with tooling for serial production to other enterprises of the branch."*
>
> — CC CPSU & USSR Council of Ministers (1968), § 18(b).
>
> **[Q-B7]** *"In the production association, scientific organizations are usually of local significance and confine their research-development-innovation activity primarily to the production needs of the association. In the NPO, on the other hand, these units are expected to conduct general-purpose or branch-wide R&D, developing innovations for the branch as a whole. The 'head' organization also differs. While this role belongs to an industrial enterprise in the production association, it is performed generally by a powerful research institute in the science-production association."*
>
> — Cocks (1980), pp. 72–73.

Operational consequence: NPO is coded Branch (head body is a research institute); pure PO without «научно» is coded Enterprise.

Coding criteria (Rules R3, R6):

Code Branch if the name contains: (i) a production-ministry tag (Министерство, Минэнерго, Минсудпром, Минсельхоз, Минздрав, Ministry); (ii) a state-committee tag (Госплан, ГКНТ, Госстрой, Госстандарт, Госкомстат, Госбанк, ЦСУ, State Committee); (iii) an NII / NPO / OKB / TsKB / SKB / KB / GSPI / VNII / TsNII tag — and the name does not satisfy a higher-priority Academy or VUZ rule. Residual rule R6: if a name still contains «институт» / 'Institute' but matches none of the more specific rules, default to Branch — per Cocks 1980 p. 57, branch is the modal destination for otherwise-unclassified Soviet research institutes.

Worked examples:

Sheet	Abbreviation	English name	Evidence
Previous Institution	Академия Коммуналь. х-ва им. Памфил…	Academy of Municipal Economy named after Pamfilov	Manual: USSR Min. Housing & Communal Economy R&D academy
Previous Institution	Всесоюз.центр переводов научно-техн…	All-Union Center for Translation of Scientific and Technical Lite…	EN: '\bState Committee\b'
Previous Institution	Всесоюз. проектно-технологич. ин-т …	All-Union Design-Technological Institute for the Mechanization of…	EN: '\bAll-Union\b.*Institute'
Previous Institution	Союзспортпроект	All-Union Institute for the Design of Sports Facilities	Manual: Soyuzsportproekt design institute
Previous Institution	ВИЖ	All-Union Institute of Animal Husbandry	EN: '\bAll-Union\b.*Institute'

4.3 VUZ (Higher Education / вузовский сектор)

Definition. The VUZ sector comprises higher educational institutions (vysshiye uchebnyye zavedeniya), their teaching and research staff, and their attached R&D infrastructure: VUZ NII, problem laboratories (проблемные лаборатории), branch laboratories (отраслевые лаборатории), scientific (research) sectors of kafedry, IPK advanced-training institutes, and tekhnikums. 85–90% of VUZ R&D operates under Minvuz (USSR Ministry of Higher and Specialized Secondary Education) or its union-republic counterparts.

Verbatim definitional evidence:

[Q-V1] *"Higher school or VUZ scientific institutes and laboratories conduct R&D in all branches of science and the humanities. The great majority of VUZ NO's, accounting for perhaps 85 to 90 percent of VUZ R&D, fall under VUZes subordinate to the U.S.S.R. Ministry of Higher and Secondary Specialized Education (Minvuz) or its union-republic counterparts."*

— Nolting (1985), p. 6.

[Q-V2] *"Institutions of higher education (VUZy), their teaching staffs, and their students constitute an important R&D resource and the third relatively independent subsystem… In total, there were more than 800 civilian higher educational institutions in the Soviet Union in 1972. Some 60 percent of the daytime students in the VUZy reportedly participate in research."*

— Cocks (1980), MinVUZ section.

[Q-V3] *"Fully one-third of the scientific workers are in the VUZy, though they account for less than five percent of all R&D performed in the country. Nonetheless, about 70 percent of all research conducted on the basis of economic contracts with industry is performed at the VUZy."*

— Cocks (1980), Ch. VIII.

[Q-V4] *"VUZ labs may be branch laboratories or problem laboratories. The former conduct research on an industrial organization's needs for new materials, processes, and equipment, whereas problem laboratories are created for the execution of major scientific, engineering, and experimental design projects."*

— Cocks (1980), p. 69.

[Q-V5] *"In VUZy under the USSR Ministry of Higher and Specialized Secondary Education at the end of 1971 there were 55 scientific research institutes, 419 problem laboratories, and 528 branch laboratories."*

— Cocks (1980), p. 69.

[Q-V6] *"At the end of 1978, VUZ NO's included 58 NII's…, 540 problem laboratories, 770 branch laboratories, and about 400 scientific research sectors."*

— Nolting & Feshbach (1981), p. 3, footnote 2.

Statistical magnitude:

Indicator	Value	Source
VUZ count 1956 → 1975	767 → 856	NF81 Table 1, p. 4
Civilian VUZes (1972)	> 800	Cocks 1980 Ch. VIII
Share of scientific workers (nominal)	≈ 33–37%	Cocks p. 46; NF80 p. 497
Share (full-time R&D equivalent)	≈ 15%	NF80 p. 497
Share of official science expenditures	5–9%	Cocks p. 46; NF80 pp. 496–497
Share of contract research with industry	≈ 70%	Cocks 1980 Ch. VIII
Year-end 1978 VUZ NO inventory	58 NII + 540 problem + 770 branch + ~400 sectors	NF81 p. 3 fn 2
Full-time researchers in VUZy: 1965 → 1976	7.7% → 17.9%	NF81 p. 28

Sub-types: Universities (МГУ, ЛГУ, …); polytechnic and higher technical schools; specialized institutes (МАДИ, МАМИ, МИРЭА, МИУ, МХТИ, МИСИ, МЛТИ, МИЭИ, МЭИ, МИЭТ, МИИТ, МАИ, МИФИ, МФТИ, МИНХ, МЭСИ, МГИМО); agricultural academies (Timiryazev, Estonian Agricultural Academy); military VUZ-equivalent academies (Zhukovsky Air Force Academy); specialized state academies (Foreign Trade, National Economy / АНХ, MVD, Diplomatic, Social Sciences / АОН при ЦК КПСС); VUZ NII; problem laboratories; branch laboratories; scientific sectors of kafedry; ИПК advanced-training institutes; tekhnikums.

Coding criteria (Rule R3 / R-VUZ):

Code VUZ if the name contains a higher-education marker: «университет», «политехнический», «институт инженеров», «педагогический», «медицинский», «финансовый», «институт народного хозяйства», «институт повышения квалификации (ИПК)», «техникум», «высшее училище», or a Moscow named-after specialist institute (МАДИ/МАМИ/МИРЭА/МИУ/МХТИ/МИСИ/МЛТИ/МИЭИ/МЭИ/МИЭТ/МИИТ/МАИ/МИФИ/МФТИ/МИНХ). Specialized state academies and military training academies are also coded VUZ here, per Q-V2 (Cocks's identification of MinVUZ as the third independent subsystem).

Worked examples:

Sheet	Abbreviation	English name	Evidence
Previous Institution	Академия МВД	Academy of the USSR Ministry of Internal Affairs	EN: '\bAcademy of the USSR Ministry of Internal Affairs\b'

Previous Institution	Академия им. Жуковского	Air Force Engineering Academy named after Professor N. E. Zhukovs…	RU: 'Военно-воздушная.*академия'
Previous Institution	Всесоюз.заоч.учетно-кредит.техник.Г…	All-Union Correspondence Accounting and Credit Technical School o…	EN: '\bSchool of\b'
Previous Institution	Всесоюз. заочн. ин-т пищевой промыш…	All-Union Correspondence Institute of Food Industry	EN: '\bCorrespondence Institute\b'
Previous Institution	ВЗИИТ	All-Union Correspondence Institute of Railway Transport Engineers	RU: 'институт инженеров'

4.4 Enterprise (заводская наука / 'plant sector of science')

Definition. The Enterprise sector comprises R&D, design, and technological subdivisions embedded in production enterprises (заводы / фабрики / комбинаты / производственные объединения / трест) and serving primarily the needs of their own production line. Per Cocks 1980, factory science has historically been small in the Soviet system — only 3% of all scientific personnel and 2% of official science expenditures are concentrated here.

Verbatim definitional evidence:

[Q-E1] *"An enterprise is a legally independent entity concerned almost exclusively with production. It has its own technical, production, and financial plan (tekhpromfinplan)… It most frequently includes a single plant. The term enterprise (predpriyatiye) is also a generic term that covers a number of forms of production organization. One is the plant (zavod)… The term factory (fabrika) is used primarily for plants in light industry… When several technologically related production activities are combined, the resulting enterprise is called a combine (kombinat)."*

— Cocks (1980), p. 70.

[Q-E2] *"In general, 'factory science' has not been a prominent feature of the Soviet industrial order. Historically, the organizational approach has emphasized the separation of industrial research from production as well as the centralization of R&D forces in institutes designed to serve the needs of the branch as a whole rather than of individual enterprises. Consequently, most enterprises lack adequate in-house R&D facilities."*

— Cocks (1980), p. 73.

[Q-E3] *"The enterprise-level R&D system of factory laboratories, design offices, experimental shops, and other scientific subdivisions serves primarily the needs of current production. In fact, enterprise scientific subdivisions are not classified under the 'science and science services sector' category of economic and social organizations, and their activity is not included in the national plan section for financing research and design work."*

— Cocks (1980), p. 73.

[Q-E4] *"There are four major types of scientific and technical subdivisions in industrial enterprises: (i) enterprise laboratories, which are the most numerous and employ the largest numbers of unspecialized personnel and engineering technical workers; (ii) design organizations; (iii) test experimental organizations; and (iv) mechanization and automation departments. Approximately 1,375,000 persons were employed in enterprise scientific and technical subdivisions in 1973."*

— Nolting & Feshbach (1980), p. 497.

[Q-E5] *"Only 3 percent of all scientific personnel and 2 percent of official science expenditures are concentrated in production enterprises and organizations."*

— Cocks (1980), p. 46.

Sub-types (verbatim from Nolting & Feshbach 1981 Appendix B § II.B, pp. 53–54):

(1) Laboratories — central plant (TsZL — центральная заводская лаборатория), shop, and plant administration laboratories.
(2) Design divisions — special testing and design bureaus (OSKB — особое специальное конструкторское бюро), departments of main designer, design bureaus, project-design departments, technological departments, design divisions of shops and plant administrations.
(3) Test-experimental departments.
(4) Departments of mechanization and automation.

Coding criteria (Rule R4 / R-Enterprise):

Code Enterprise if the name contains: «завод», «фабрика», «комбинат», «производственное объединение» (without «научно»), «трест», «автокомбинат», «отд. ж.д.», 'Factory', 'Plant', 'Combine', 'Railway Branch', 'Railway Station', 'Trust', 'Depot', 'Automobile Transport Complex'.

Worked examples:

Sheet	Abbreviation	English name	Evidence
Previous Institution	Трест Центроэнергомонтаж	Central Power Engineering Installation Trust	Manual: Tsentroenergomontazh trust
Previous Institution	ЦТМП	Central Topographic and Mine Surveying Enterprise	Manual: TsTMP enterprise
Previous Institution	З-д экспериментального машиностроен…	Experimental Machine-Building Plant	EN: '\bPlant\b'
Previous Institution	Завод счетно-аналитических машин	Factory for the Production of Accounting and Analytical Machines	EN: '\bFactory\b'
Previous Institution	Завод ВТУЗ при Автозаводе им.Лихаче…	Factory-Based Higher Technical Educational Institution at the Mos…	RU: 'завод(?!ская)'

4.5 Other (non-R&D / outside the 4-sector framework)

Definition. The Other code is reserved for bodies that fall outside the TsSU 'science and science services' branch by explicit definition. We also include here closed institutions (military units в/ч, postal-box п/я entities) whose actual sector cannot be determined from the name string alone.

Verbatim exclusion evidence:

> **[Q-O1]** *"The following types of organizations, though classified as scientific institutions, are not included in the science and science services branch: higher educational institutions (vysshiye uchebnyye zavedeniya — VUZy), except for teacher training schools; separate NO's under the administration of the VUZY, with the exception of those NO's classified as NII's; and scientific and technical administrative and planning offices of state supervisory agencies and branch ministries… Also excluded from the science and science services branch are the scientific, technical, and experimental laboratories, departments, shops, and plants of industrial enterprises…, except those which are officially designated as 'scientific institutions' because of their size and importance."*
>
> — Nolting & Feshbach (1981), p. 3.
>
> **[Q-O2]** *"Scientific organizations not classified as scientific institutions include nonresearch design organizations, experimental plants that do not produce industrial products for sale, maritime resources prospecting organizations, industrial testing laboratories, computer and information centers not engaged in research, hydrometeorological organizations, and geological survey and prospecting organizations."*
>
> — Nolting & Feshbach (1980), p. 494.
>
> **[Q-O3]** *"The most detailed breakdown of the residual was published for 1970, in which the proportion of the residual to total scientific workers was given as 7.97 percent, of which industrial enterprises employed 3.27 percent, nonresearch design and project organizations 3.16 percent, and central administrative departments and other organizations 1.54 percent."*
>
> — Nolting & Feshbach (1981), p. 32.

Worked examples:

Sheet	Abbreviation	English name	Evidence
Previous Institution	АРЕ г. Александрия	Alexandria, Egypt	RU: '^АРЕ'
Previous Institution	ВТБИЛ	VTBIL	Manual: VGBIL — Library of Foreign Literature
Previous Institution	Алтайский госзаповедник	Altai State Nature Reserve	RU: 'заповедник'
Previous Institution	ЦК профсоюза работн. с/х	Central Committee of the Trade	RU: 'профсоюз'

		Union of Agricultural Workers	
Previous Institution	Центр. ин-т повышения квалификации …	Central Institute for Advanced Training of Managerial Personnel i…	RU: 'кадров'

5. Russian-Language R&D Vocabulary Glossary

Acronym	Cyrillic	English	Verbatim definition + source
НИИ (NII)	научно-исследовательский институт	Scientific Research Institute	"NIIs form the bulk of the research organizations of the Academy system and are the most prestigious R&D performers of all Soviet R&D organizations." — Kassel & Campbell (1980), RAND R-2533, pp. 10–11.
ЦНИИ (TsNII)	центральный научно-исследовательский институт	Central / Head Scientific Research Institute	Head institute of a branch. — Cocks (1980), Ch. VIII, p. 65.
ВНИИ (VNII)	всесоюзный научно-исследовательский институт	All-Union Scientific Research Institute	All-union NII serving the entire USSR within a branch. — NF81 p. 3, App. B.
ОКБ (OKB)	опытно-конструкторское бюро	Test/Experimental Design Bureau	"test-design bureaus (OKB)." — Nolting (1985), Ch. 1, p. 4.
ЦКБ (TsKB)	центральное конструкторское бюро	Central Design Bureau	"central design bureaus (TsKB)." — Nolting (1985), Ch. 1, p. 4.
СКБ (SKB)	специальное конструкторское бюро	Special Design Bureau	"special design bureaus (SKB / OKB)." — Nolting (1985), Ch. 1, p. 4.
КБ (KB)	конструкторское бюро	Design Bureau	"The Design Bureau or KB, found mainly in the industrial ministry system, engages in the middle and late stages of RDI, specializing in product design." — Kassel & Campbell (1980), RAND R-2533, p. 10.
ГСПИ (GSPI)	государственный союзный проектный институт	State Union Project Institute	"project institutes that specialize in the designing and planning of new plants or renovation of old enterprises." — Cocks (1980), Ch. VIII, p. 65.
НПО (NPO)	научно-производственное объединение	Scientific-Production Association	"scientific–production associations, comprising research

			institutes with design–engineering and technological divisions and industrial enterprises." — Decree No. 760 of 24 September 1968, § 18(b).
ПО (PO)	производственное объединение	Production Association	"By the fall of 1976 there were more than 3000 POs in industry." — Cocks (1980), Ch. VIII, p. 71.
ВУЗ (VUZ)	высшее учебное заведение	Higher Educational Institution	"higher educational institutions (vysshiye uchebnyye zavedeniya — VUZy)." — Nolting & Feshbach (1981), p. 3.
ИПК (IPK)	институт повышения квалификации	Institute of Advanced Training	Part of the VUZ sector taxonomy in NF81 Appendix B § II.A and Nolting 1985 Ch. 1; treated as VUZ-equivalent in coding rule R3.
АН СССР (AN SSSR)	Академия наук СССР	USSR Academy of Sciences	"U.S.S.R. Academy of Sciences, which conducts planning and coordination for the entire academy system." — Nolting & Feshbach (1981), p. 27.
Главки	главные управления	Main Administrations / Chief Directorates	"grouped by product line and subordinated to units of their respective ministries known as chief directorates or main administrations (Glavki)." — Cocks (1980), Ch. VIII, p. 58.
Отраслевая наука	отраслевая наука	Branch Science	"'branch science'… is commonly used in referring to this tripartite division." — Cocks (1980), Ch. VIII, pp. 45–46. Operational definition: NF81 pp. 25–26.
Заводская наука	заводская наука	Factory / Plant Science	"Enterprise S&T departments… are frequently referred to as the 'plant sector of science'." — Nolting (1985), Ch. 1, pp. 6–7.

Академический сектор	академический сектор	Academic Sector	Headlined as the Academy sector in Nolting (1985), Ch. 1, p. 5; corresponds to Cocks's 'academy science' (1980, Ch. VIII, p. 46).

6. Coding Workflow — Cascade Rules R0–R7

The coding algorithm applies the following rules in strict order. The first matching rule wins. Each rule is anchored in one or more verbatim quotes from §§ 2 and 4 above.

Rule	Trigger	Action	Anchored in
R0	Manual override	If the institution appears in the manual-override dictionary (e.g., 'Academy of Municipal Economy named after Pamfilov' → Branch; 'Joint Institute for Nuclear Research' → Branch; 'Moscow Agricultural Academy named after Timiryazev' → VUZ; 'CEMI' / Estonian or Leningrad Branch of CEMI → Academy), apply that decision and stop.	Coder's discretion based on Q-A1 … Q-V6
R1	Non-R&D filter	If the name contains explicit non-R&D markers — library, publishing house, journal, museum, hospital, sanatorium, sports society, post office, secondary school, military unit (в/ч), postal box (п/я), foreign organization — code Other and stop.	Q-O1, Q-O2 (NF81 p. 3; NF80 p. 494)
R2	Academy marker	If the name contains «АН», «Академия наук», «АМН», «ВАСХНИЛ», «АПН» (or English equivalent), code Academy. Override exceptions: military training academies, specialized state academies, Timiryazev → VUZ; Pamfilov Municipal Economy → Branch.	Q-A1, Q-A2 (NF81 p. 27; Cocks p. 47)
R3	VUZ marker	If the name contains a higher-education marker (university, polytechnic, pedagogical, medical, financial, mining, institute of national economy, IPK, tekhnikum, higher school, Moscow named-after specialist institute), code VUZ.	Q-V1, Q-V2 (Nolting 1985 p. 6; Cocks Ch. VIII)
R4	Branch marker	If the name contains a production-ministry tag, a state-committee tag, or an NII/NPO/OKB/TsKB/SKB/KB/GSPI/VNII/TsNII tag — and the name does not satisfy R2 or R3 — code Branch.	Q-B1, Q-B2 (NF81 pp. 25–26; Cocks p. 57)
R5	Enterprise marker	If the name contains «завод», «фабрика», «комбинат», «производственное объединение»	Q-E1, Q-E4 (Cocks p. 70; NF80 p. 497)

		(without «научно»), «трест», «автокомбинат», «отд. ж.д.», or English equivalent, code Enterprise.	
R6	Residual Institute → Branch	If the name still contains «институт» / 'Institute' but matches none of the more specific rules, default to Branch — per Cocks 1980 p. 57, branch is the modal destination for otherwise-unclassified Soviet research institutes.	Q-B2 (Cocks 1980 p. 57)
R7	Final residual	Otherwise → Other.	Q-O3 (NF81 p. 32 residual)

7. Data Quality Caveats and Treatment

7.1 Defense / military exclusion

> **[Q-Q1]** *"The undesignated residual employment category (labeled other branches in Soviet sources) given in Soviet statistics apparently consists of military or military-related sciences not otherwise included in the technical sciences… To our knowledge, statistics have never been published on the distribution of scientific workers in military R&D, let alone by specific branch."*
>
> — Nolting & Feshbach (1980), pp. 498–499.

Treatment. Military units (в/ч) and postal-box (п/я) entities — which masked closed facilities frequently in defense industry — are coded Other rather than guessed at Branch or Enterprise. This sacrifices a small number of likely-Branch coding events for transparency about what cannot be confidently determined.

7.2 Nomenclature heterogeneity

> **[Q-Q2]** *"Today there are over 100 designations of scientific organizations in the USSR. However, the correspondence between designation and function is generally poor. One American authority observes: 'The nomenclature of scientific organizations has become for the most part a hodge-podge rather than an indication of functional type.'"*
>
> — Cocks (1980), p. 64, citing Nolting (1976).

Treatment. No single keyword is treated as decisive — the cascade rules R0–R7 use multiple intersecting markers (ministry tag + institute-type tag + named-after pattern) and a manual-override list resolves the most heterogeneous cases.

7.3 Definition shifts over time

> **[Q-Q3]** *"The 1961 transfer of some Academy institutes to industrial jurisdiction failed; neither extreme — full industrialization of the Academy or full merger into the ministries — is likely, and the Soviet system will continue to 'patch rather than rebuild' the configuration."*
>
> — Kassel & Campbell (1980), p. 25.

Treatment. We code on the affiliation reported in the source name string (the affiliation of record at the time of the CEMI personnel event); the 1961 transfer is not retroactively applied.

7.4 Enterprise / Branch boundary blurred by reclassification

> **[Q-Q4]** *"Many — particularly large — central plant laboratories (центральные заводские лаборатории) are officially designated as scientific institutions, and their scientific workers are statistically recorded as employees of the science and science services sector rather than of industry."*

— Nolting & Feshbach (1980), p. 498.

Treatment. A central plant laboratory (TsZL) embedded in a production enterprise is coded Enterprise by default; if it has been formally upgraded to scientific-institution status under a ministry, the affiliation string will normally reflect that and Rule R4 (Branch) will fire first.

8. Per-Sheet Distribution & Interpretation

Sheet	Academy	Branch	VUZ	Enterprise	Other	Total
Sheet1 Previous Institution	66	247	68	25	39	445
Sheet2 Part-Time Institution	1	13	40	2	12	68
Sheet3 Transferred Institution	16	70	5	9	21	121
TOTAL	83	330	113	36	72	634
Share (%)	13.1	52.1	17.8	5.7	11.4	100.0

Interpretation. Sheet1 (Previous Institution, Branch 55%) directly reflects Nolting & Feshbach's (1981, pp. 25–26) observation that branch institutions employed nearly half of all Soviet scientific workers and concentrated the bulk of applied R&D. Academy share (15%) is structurally higher than the all-USSR figure (9%) because CEMI itself is an Academy institute and disproportionately recruits researchers from the academic sector. Sheet2 (Part-Time, VUZ 59%) reflects the well-attested pattern of CEMI researchers holding teaching part-time appointments at Moscow VUZes — a continuation of the Soviet practice of integrating academic researchers into university teaching (Q-V2: "some 60 percent of the daytime students in the VUZy reportedly participate in research"). Sheet3 (Transferred, Branch 58%) is consistent with the outflow of trained economists from CEMI into ministry-affiliated NII-type institutes (Gosplan, GKNT, Goskomstat NIIs) — branch is the modal destination of mobile scientific workers (Q-B5: 78.6% growth in branch institutions 1956–1975).

9. Worked Coding Examples

Five examples per sector — actual entries from the classified glossary. The Evidence column shows the matching keyword or manual-override rationale.

9.1 Academy

Sheet	Abbreviation	English name	Evidence
Previous Institution	АН Азербайджан. ССР	Academy of Sciences of Azerbaijan SSR	RU: 'АН '
Previous Institution	АН Казахской ССР	Academy of Sciences of Kazakh SSR	RU: 'АН '
Previous Institution	АН Литовской ССР	Academy of Sciences of Lithuanian SSR	RU: 'АН '
Previous Institution	АН Молдавской ССР	Academy of Sciences of Moldavian SSR	RU: 'АН '
Previous Institution	АН Таджиковой ССР	Academy of Sciences of Tajik SSR	RU: 'АН '

9.2 Branch

Sheet	Abbreviation	English name	Evidence
Previous Institution	Академия Коммуналь. х-ва им. Памфил…	Academy of Municipal Economy named after Pamfilov	Manual: USSR Min. Housing & Communal Economy R&D academy
Previous Institution	Всесоюз.центр переводов научно-техн…	All-Union Center for Translation of Scientific and Technical Lite…	EN: '\bState Committee\b'
Previous Institution	Всесоюз. проектно-технологич. ин-т …	All-Union Design-Technological Institute for the Mechanization of…	EN: '\bAll-Union\b.*Institute'
Previous Institution	Союзспортпроект	All-Union Institute for the Design of Sports Facilities	Manual: Soyuzsportproekt design institute
Previous Institution	ВИЖ	All-Union Institute of Animal Husbandry	EN: '\bAll-Union\b.*Institute'

9.3 VUZ

Sheet	Abbreviation	English name	Evidence
Previous Institution	Академия МВД	Academy of the USSR Ministry of Internal Affairs	EN: '\bAcademy of the USSR Ministry of Internal Affairs\b'
Previous Institution	Академия им. Жуковского	Air Force Engineering Academy named after Professor N. E. Zhukovs…	RU: 'Военно-воздушная.*академия'
Previous Institution	Всесоюз.заоч.учетно-кредит.техник.Г…	All-Union Correspondence Accounting and Credit Technical School o…	EN: '\bSchool of\b'
Previous Institution	Всесоюз. заочн. ин-т пищевой промыш…	All-Union Correspondence Institute of Food Industry	EN: '\bCorrespondence Institute\b'
Previous Institution	ВЗИИТ	All-Union Correspondence Institute of Railway Transport Engineers	RU: 'институт инженеров'

9.4 Enterprise

Sheet	Abbreviation	English name	Evidence
Previous Institution	Трест Центроэнергомонтаж	Central Power Engineering Installation Trust	Manual: Tsentroenergomontazh trust
Previous Institution	ЦТМП	Central Topographic and Mine Surveying Enterprise	Manual: TsTMP enterprise
Previous Institution	З-д экспериментального машиностроен…	Experimental Machine-Building Plant	EN: '\bPlant\b'
Previous Institution	Завод счетно-аналитических машин	Factory for the Production of Accounting and Analytical Machines	EN: '\bFactory\b'
Previous Institution	Завод ВТУЗ при Автозаводе им.Лихаче…	Factory-Based Higher Technical Educational Institution at the Mos…	RU: 'завод(?!ская)'

9.5 Other

Sheet	Abbreviation	English name	Evidence
Previous Institution	АРЕ г. Александрия	Alexandria, Egypt	RU: '^АРЕ'
Previous Institution	ВТБИЛ	VTBIL	Manual: VGBIL — Library of Foreign Literature
Previous Institution	Алтайский госзаповедник	Altai State Nature Reserve	RU: 'заповедник'
Previous Institution	ЦК профсоюза работн. с/х	Central Committee of the Trade Union of Agricultural Workers	RU: 'профсоюз'
Previous Institution	Центр. ин-т повышения квалификации …	Central Institute for Advanced Training of Managerial Personnel i…	RU: 'кадров'

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Appendix A: Per-Rule Citation Index

Rule	Anchoring Q-IDs and citations
R0 Manual override	Coder discretion based on Q-A1 … Q-V6
R1 Non-R&D filter	Q-O1 NF81 p. 3 (exclusion list); Q-O2 NF80 p. 494
R2 Academy marker	Q-A1 NF81 Ch. 3 p. 27; Q-A2 Cocks p. 47; Q-A3 Cocks pp. 48–49; Q-A4 Cocks p. 50; Q-A6 Kassel & Campbell p. 20
R3 VUZ marker	Q-V1 Nolting 1985 p. 6; Q-V2 Cocks Ch. VIII; Q-V3 Cocks; Q-V4 Cocks p. 69; Q-V5 Cocks p. 69; Q-V6 NF81 p. 3 fn 2
R4 Branch marker	Q-B1 NF81 pp. 25–26; Q-B2 Cocks p. 57; Q-B3 Cocks p. 57; Q-B4 Cocks p. 58; Q-B5 NF81 Table 1 p. 4; Q-B6 Decree 760/1968 § 18(b); Q-B7 Cocks pp. 72–73
R5 Enterprise marker	Q-E1 Cocks p. 70; Q-E2 Cocks p. 73; Q-E3 Cocks p. 73; Q-E4 NF80 p. 497; Q-E5 Cocks p. 46
R6 Residual Institute → Branch	Q-B2 Cocks p. 57
R7 Final residual	Q-O3 NF81 p. 32